Hands-On-Learning: Experiment Orchestrator – From Data to Decision

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Learning Objectives

By the end of this activity, learners will be able to:

- Run A/B/C experiments using a reproducible evaluation harness.
- Compute latency percentiles (p50/p95), tokens/sec, cost per task, and win-rate.
- Visualize model or prompt performance to compare variants effectively.
- Make a structured Ship/No-Ship decision with justification and rollback criteria.

Description

In this hands-on practice, you will run controlled experiments across three LLM configurations, compute meaningful performance metrics, visualize outcomes, and produce a decision memo—mirroring the workflow AI product teams use before promoting new model variants.

Structure of the Activity

Title

Experiment Orchestrator – From Data to Decision

Dataset Information / Resources Required

- **Test dataset:** JSONL or CSV with 20–30 evaluation cases
- **Starter harness:** Python evaluation script template
- **Logging templates:**
  - results/variant_A.csv
  - results/variant_B.csv
  - results/variant_C.csv

Tools Used

- Python 3.10+
- OpenAI API (or compatible SDK)
- pandas
- matplotlib
- JSON/CSV evaluation dataset

Setup Requirements

- Install required libraries:
  pip install openai pandas matplotlib
- Store your API key in an environment variable
- Place the dataset in a data/ folder
- Ensure the starter harness runs correctly in your environment

Scenario

Your team has tested three LLM configurations for a productivity assistant and now needs a data-driven recommendation on which variant to ship. You will run identical evaluations across all variants, compare performance, and provide a Ship/No-Ship recommendation supported by metrics and a rollback criterion.

Objective

By the end of this exercise, you will produce a complete experiment comparison pack: metrics tables, a visualization, and a short decision memo recommending which model or prompt variant should go live.

Instructions for Participants

Getting Started

- Open the starter evaluation harness.
- Load your dataset (20–30 evaluation cases).
- Confirm that variants A/B/C use consistent settings and seeds.
- Create a results/ folder for output CSVs and charts.

Activity 1: Run A/B/C Experiments

Objective

Execute the harness for three variants

and collect clean, comparable metrics.

Input/Data

Dataset: data/eval_cases.jsonl or .csv

Outputs: 3 CSVs (latency, tokens, quality scores, cost)

Steps to Complete

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Step 1: Run Variant A

Use the baseline configuration

Export to results/variant_A.csv

Step 2: Run Variant B

Use the “fast” configuration

Export to results/variant_B.csv

Step 3: Run Variant C

- Use the “high-quality” configuration
- Export to results/variant_C.csv

Step 4: Compute Key Metrics (pandas)

- p50 latency
- p95 latency
- tokens per second

cost per task

win-rate vs Variant A

Step 5: Create a Combined Metrics Table

Merge A/B/C results

Compute aggregated metrics

Save as results/summary_metrics.csv

Expected Outcome

Three CSV files (A/B/C) and one summary table comparing all variants.

Tips

- Keep seeds and inputs identical across variants.
- Outliers affect p95 more than p50.
- Win-rate evaluates quality, not speed.

Activity 2: Visualize & Interpret Results

Objective

Create a bar or line chart that highlights differences clearly.

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Input/Data

Aggregated DataFrame from Activity 1

Steps to Complete

Step 1: Select a metric (p95 latency, cost/task, win-rate)

Step 2: Create a matplotlib chart, e.g.:

```
import matplotlib.pyplot as plt

plt.bar(['A','B','C'], [p95_A, p95_B, p95_C])
plt.title('p95 Latency Comparison')
plt.ylabel('Milliseconds')
plt.savefig('results/p95_latency.png')
```

Step 3: Save the chart to results/

Step 4: Interpret results (fastest, cheapest, best win-rate, trade-offs)

Expected Outcome

A clear visual comparison among the three variants.

Tips

Use one metric per chart.

Choose the metric that best clarifies the decision.

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Compile Your Findings

1 Metrics Table (CSV)

Requirement: summary_metrics.csv containing:

- p50 / p95 latency
- tokens/sec
- cost/task
- win-rate vs baseline

2 Visualization Chart

Requirement: bar/line chart

Formats: .png, .jpg, .pdf, .svg, .mp4

3 Decision Memo

Include:

- Your recommended variant
- Justification (metrics + trade-offs)
- Rollback trigger
- Next-step improvement suggestion

Summary

This activity provides hands-on experience with LLM experiment orchestration through practical application of A/B/C benchmarking, metric analysis, and structured decision-making. Participants have developed competency in running controlled evaluations, interpreting performance differences, and producing evidence-based Ship/No-Ship recommendations.

Key takeaways:

- Reliable decisions require comparing variants using meaningful metrics.
- Tail behavior (p95) exposes issues hidden by averages.
- Clear visualizations and rollback criteria strengthen production readiness.

Resources

Primary Materials:

- Python evaluation harness template
- Test dataset (JSONL + CSV)
- Logging templates (variant_A.csv, variant_B.csv, variant_C.csv)

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Reference Documentation:

- OpenAI Evaluation & Math for Developers
- pandas + matplotlib docs

Additional Resource (Optional):

- Example “Ship/No-Ship” memo structure
- Example matplotlib templates

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.